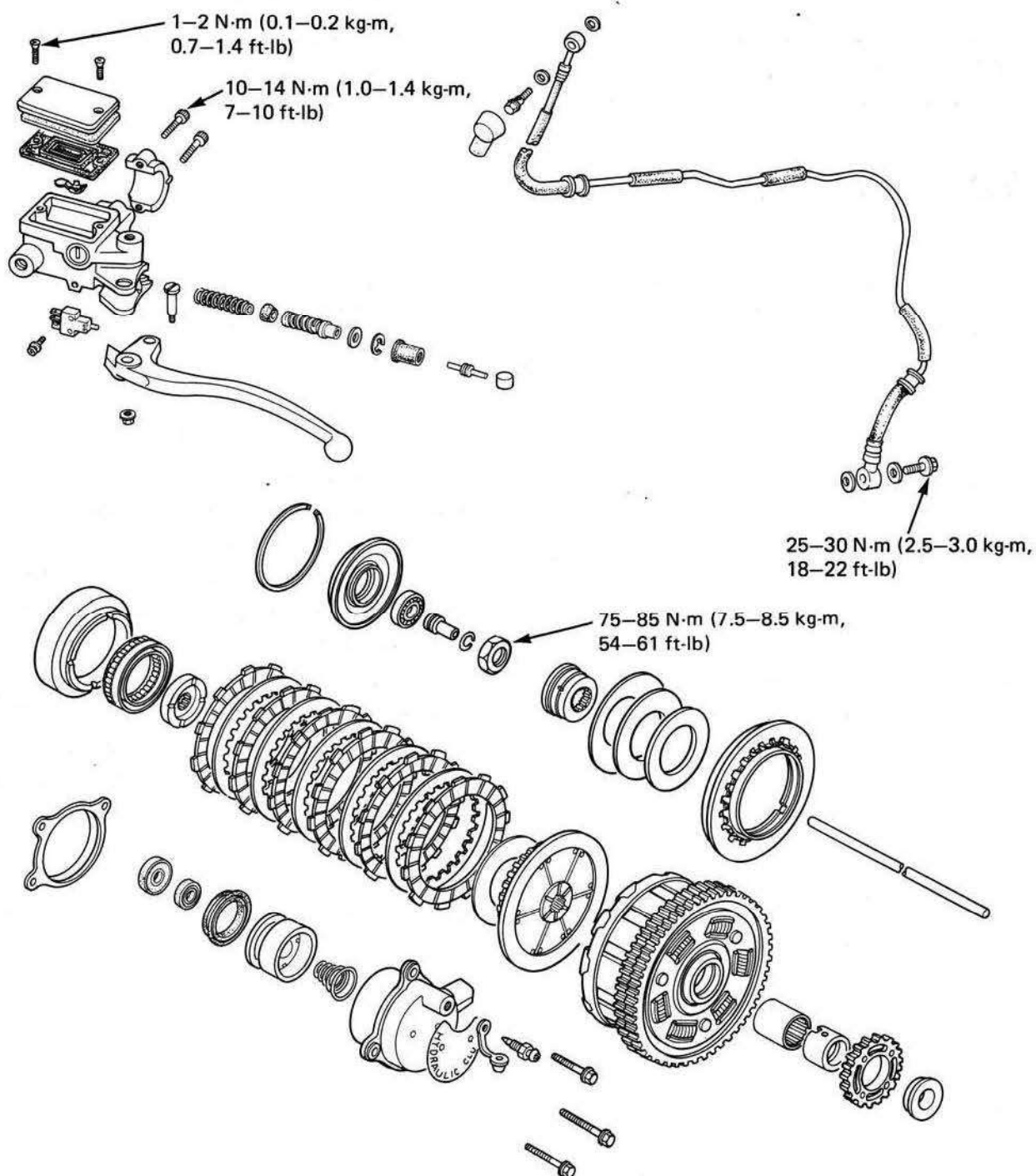




HONDA CBX750F

8. Clutch





SERVICE INFORMATION	8-1	CLUTCH SLAVE CYLINDER	8-7
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CLUTCH FLUID REPLACEMENT & AIR BLEEDING	8-3	CLUTCH ASSEMBLY	8-14
CLUTCH MASTER CYLINDER	8-4		

SERVICE INFORMATION

GENERAL

- This section covers removal and installation of the clutch hydraulic system and clutch
- DOT 4 brake fluid is used for the hydraulic clutch and is referred to as clutch fluid in this section. Do not use other types of fluid as they are not compatible.
- Clutch maintenance can be done with the engine in the frame.

SPECIFICATIONS

		STANDARD	SERVICE LIMIT
Clutch master cylinder	Cylinder I.D.	15.870-15.913 mm (0.6248-0.6265 in)	15.93 mm (0.627 in)
	Piston O.D.	15.827-15.854 mm (0.6231-0.6242 in)	15.80 mm (0.622 in)
Clutch slave cylinder	Cylinder I.D.	38.100 -38.162 mm (1.5000-1.5024 in)	38.18 mm (1.503 in)
	Piston O.D.	38.036-38.075 mm (1.4975-1.4990 in)	38.02 mm (1.497 in)
Clutch	Outside guide I.D.	24.995-25.012 mm (0.9841-0.9847 in)	25.08 mm (0.987 in)
	Spring free height	4.00 mm (0.16 in)	3.9 mm (0.15 in)
	Clutch centre B.I.D.	74.414-74.440 mm (2.9297-2.2743 in)	74.50 mm (2.933 in)
	One way clutch inner I.D.	57.755-57.768 mm (2.2738-2.2743 in)	57.74 mm (2.273 in)
	Disc thickness	3.72-3.88 mm (0.147-0.153 in)	3.1 mm (0.12 in)
	Plate warpage		0.30 mm (0.012 in)

TORQUE VALUES

Master cylinder reservoir cap	1-2 Nm (0.1-0.2 kg.m, 0.7-1.4 ft.lb)
Master cylinder holder	10-14 Nm (1.0-1.4 kg.m, 7-10 ft.lb)
Oil bolt	25-30 Nm (2.5-3.0 kg.m, 18-22 ft.lb)
Clutch lock nut	75-85 Nm (7.5-8.5 kg.m, 54-61 ft.lb)

TOOLS

Special

Snap ring pliers	07914-3230001
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Common

Extension bar	07716-0020500
Lock nut wrench, 17 x 27 mm	07716-0020300
Driver	07749-0010000
Attachment, 37 x 40 mm	07746-0010200
Pilot, 35 mm	07746-0040800
Universal holder	07725-0030000



TROUBLESHOOTING

Clutch lever soft or spongy

1. Air bubbles in hydraulic system
2. Low fluid level
3. Hydraulic system leaking

Clutch lever too hard

1. Sticking piston(s)
2. Clogged hydraulic system

Clutch slips

1. Hydraulic system sticking
2. Discs worn
3. Springs weak

Clutch will not disengage

1. Air bubbles in hydraulic system
2. Low fluid level
3. Hydraulic system leaking
4. Hydraulic system sticking
5. Plates warped

Motorcycle creeps with clutch disengaged

1. Air bubbles in hydraulic system
2. Low fluid level
3. Hydraulic system leaking
4. Hydraulic system sticking
5. Plates warped

Excessive lever pressure

1. Hydraulic system sticking
2. Lifter mechanism damaged

Clutch operation feels rough

1. Outer drum slots rough
2. Sticking piston(s)



CLUTCH FLUID REPLACEMENT/ AIR BLEEDING

Check the fluid level with the fluid reservoir parallel to the ground.

CAUTION

- Do not allow foreign material to enter the clutch system when filling the reservoir.
- Avoid spilling fluid on painted surfaces. Place a rag over the fuel tank whenever the system is serviced.

CLUTCH FLUID DRAINING

Remove the reservoir cap and diaphragm. Connect a bleed hose to the bleed valve. Loosen the slave cylinder bleed valve and pump the clutch lever. Stop operating the lever when no fluid flows out of the bleed valve.

CLUTCH FLUID FILLING

NOTE:

Do not mix different types of fluid since they are not compatible.

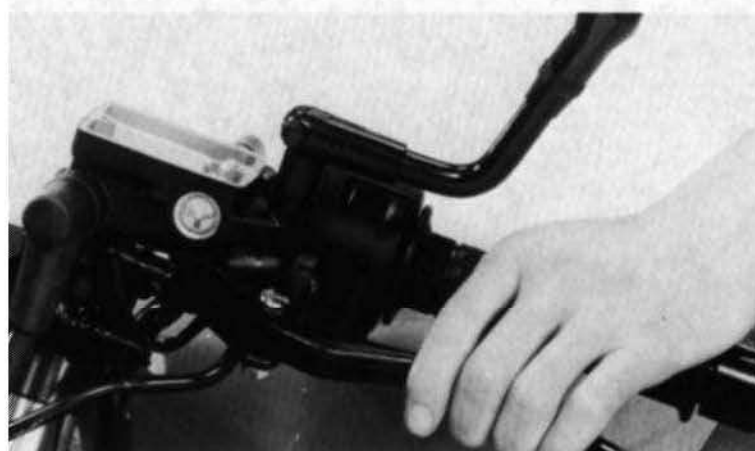
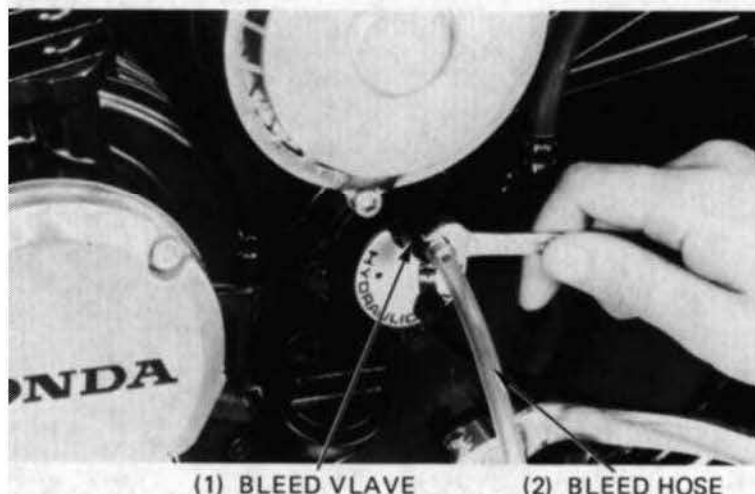
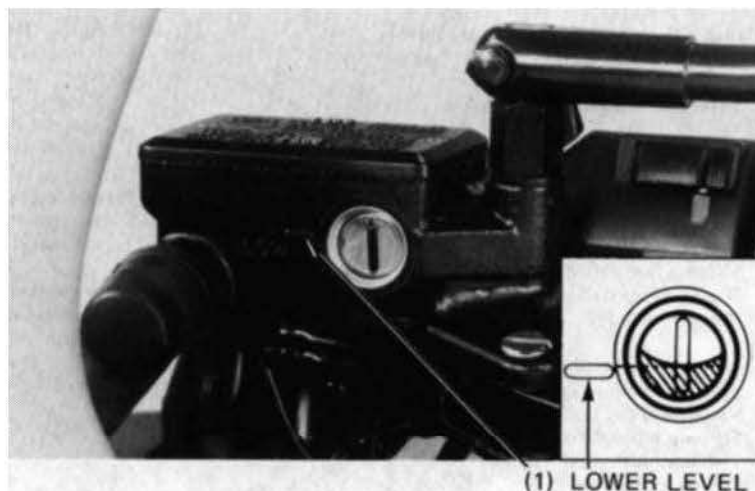
Close the bleed valve and fill the reservoir. Pump up the system pressure with the lever until there are no air bubbles in the fluid flowing out of the reservoir small hole and lever resistance is felt.

Then bleed the system (below).

AIR BLEEDING

NOTE:

- Check the fluid level often while bleeding the clutch to prevent air from being pumped into the system.
- Use only DOT 4 brake fluid from a sealed container.
- Do not mix brake fluid types and never reuse the fluid which has been pumped out during bleeding,
- or the efficiency of the clutch system will be impaired.





1) Squeeze the clutch lever, open the bleed valve 1/2 turn then close the valve.

NOTE:

Do not release the clutch lever until the bleed valve has been closed again.

2) Release the clutch lever slowly and wait several seconds after it reaches the end of its travel.

Repeat the above steps until bubbles cease to appear in the fluid at the end of the hose. Fill the fluid reservoir to the upper level.

CLUTCH MASTER CYLINDER

DISASSEMBLY

Drain clutch fluid from the hydraulic system. Remove the rear view mirror and clutch lever.

Disconnect the clutch switch wires and remove the clutch hose.

CAUTION

Avoid spilling clutch fluid on painted surfaces.

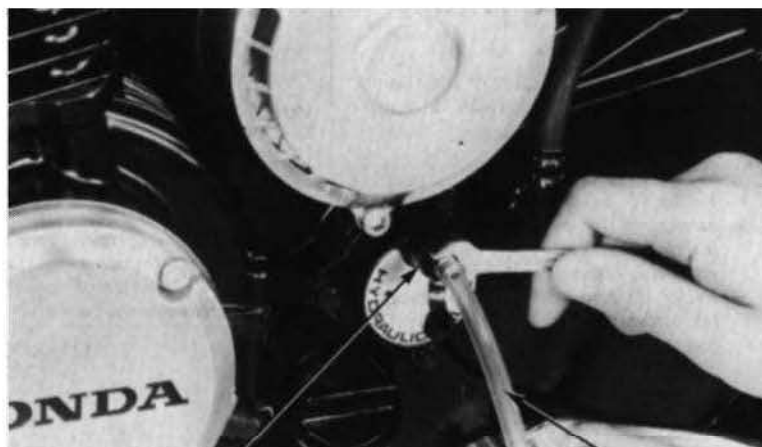
Place a rag over the fuel tank whenever the clutch system is serviced.

NOTE:

When removing the oil bolt, cover the end of the hose to prevent contamination and secure the hose.

Remove the master cylinder.

Remove the push rod, boot, and snap ring from the master cylinder body.



(1) BLEED VALVE

(2) BLEED HOSE



(1) OIL BOLT

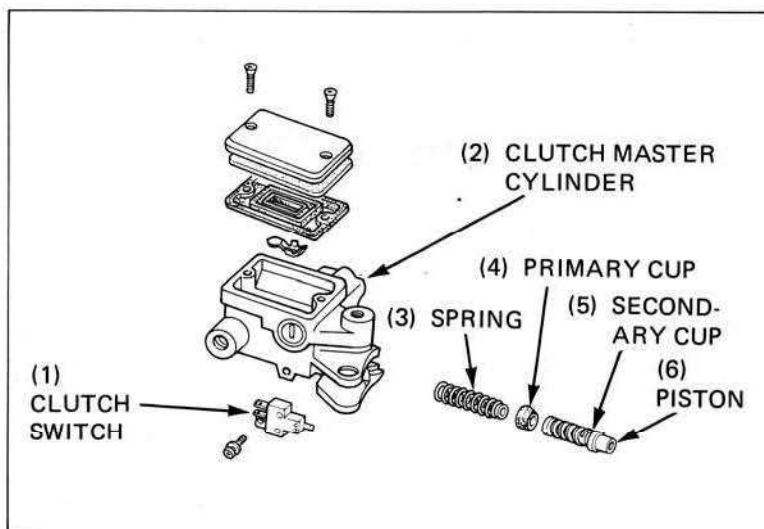
(2) CLUTCH HOSE





8. Clutch

Remove the piston, secondary cup, primary cup and spring.
Remove the clutch switch, if necessary.

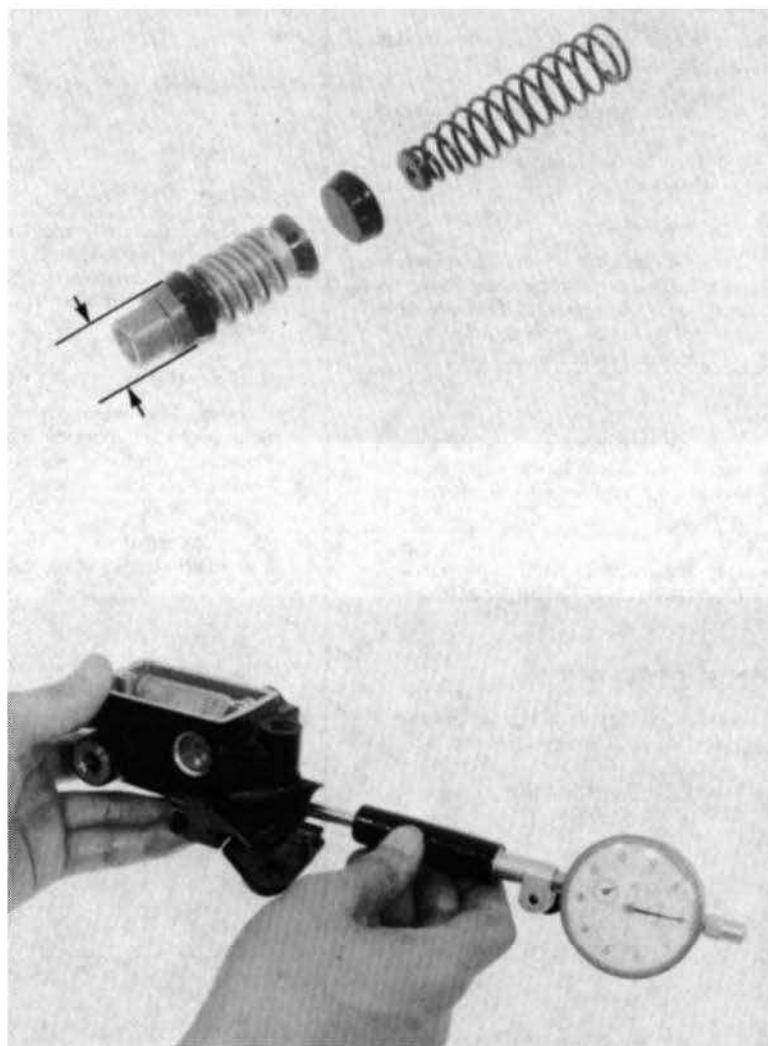


MASTER PISTON O.D. INSPECTION

Measure the master piston O.D.

SERVICE LIMIT: 15.80 mm (0.622 in)

Check the primary and secondary cups for damage before assembly.



MASTER CYLINDER I.D. INSPECTION

Check the master cylinder for scores, scratches or nicks.

Measure the master cylinder I.D.

SERVICE LIMIT: 15.93 mm (0.627 in)



ASSEMBLY

CAUTION

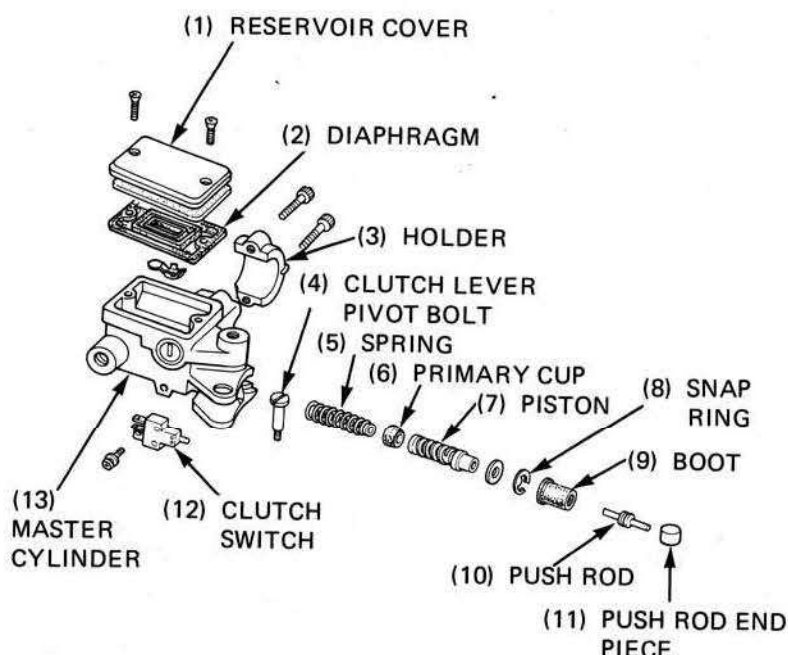
Handle the master piston, spring, primary, cup and secondary, cup as set.

Coat the primary cup and piston with clean brake fluid before assembly.
Install the spring, primary cup and piston.

CAUTION

When installing the cups, do not allow the lips to turn inside out.

Install the snap ring making sure it is seated firmly in the groove.
Install the boot and push rod.
Install the clutch switch, if it was removed.



Place the master cylinder on the handlebar and install the holder with the "UP" mark facing up and the mounting bolts.
Align the index mark on the holder with the punch mark on the handlebar, and tighten the upper bolt first, then lower bolt.

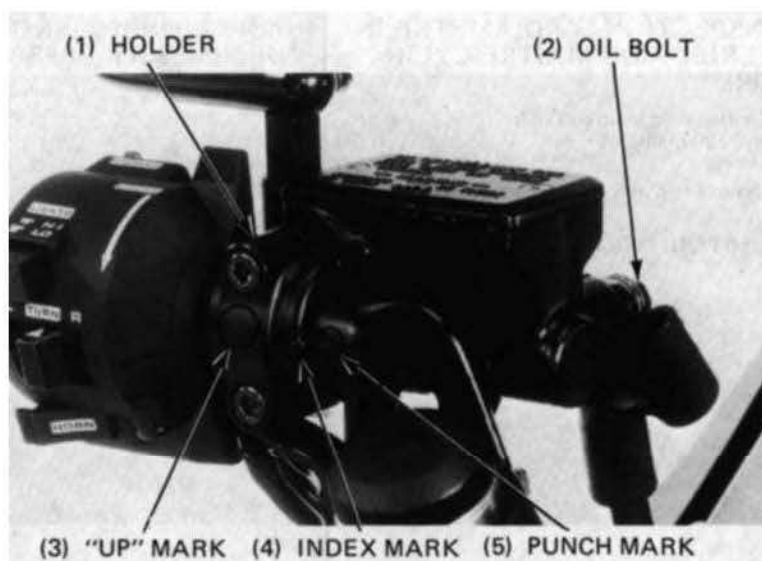
TORQUE: 10-14 Nm
(1.0-1.4 kg.m, 7-10 ft.lb)

Install the clutch hose with the oil bolt and two sealing washers, and tighten the oil bolt.

TORQUE: 25-30 Nm
(2.5-3.0 kg.m, 18-22 ft.lb)

Install the push rod end piece into the clutch lever hole and install the clutch lever.
Connect the clutch switch wires to the switch terminals.

Fill the reservoir and bleed the clutch system (page 7-3).





CLUTCH SLAVE CYLINDER

DISASSEMBLY

Place a container under the slave cylinder, remove the oil bolt and disconnect the clutch hose.

NOTE:

Avoid spilling clutch fluid on painted surfaces.

Remove the slave cylinder.

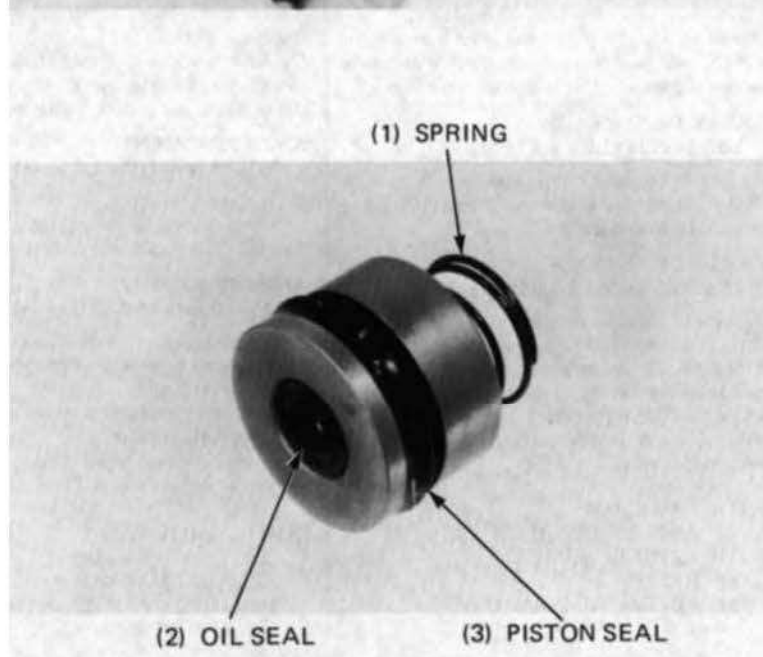
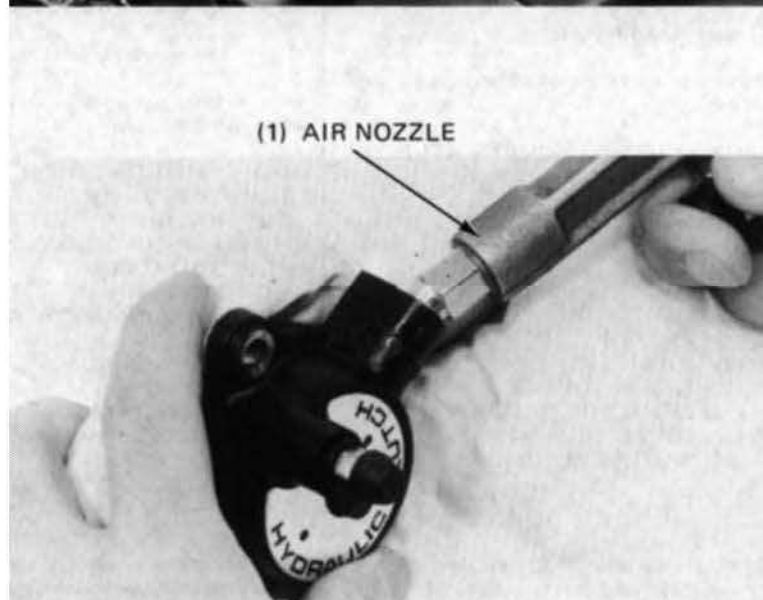
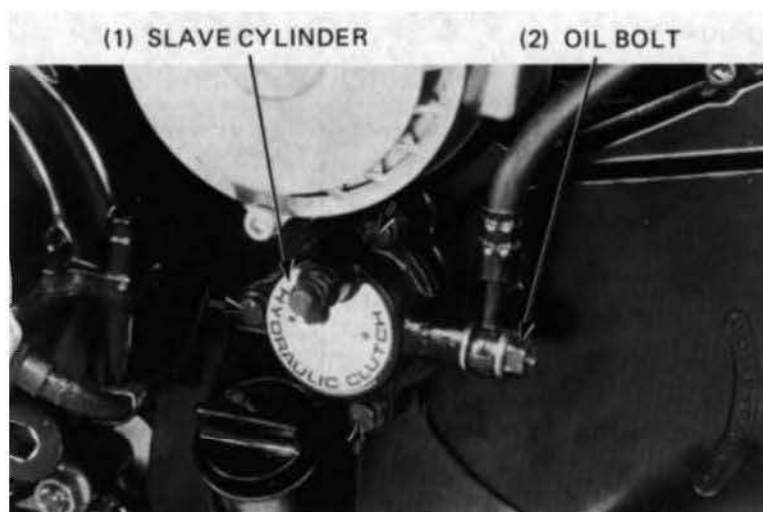
Remove the piston from the cylinder. If piston removal is hard, place a shop towel over the piston to cushion the piston when it is expelled, and position the cylinder with the piston down.

Apply compressed air to the fluid inlet to remove the piston. Use the air in short spurts.

Remove the spring from the slave cylinder.

Remove the oil and piston seals.

Clean the piston groove with clutch fluid. Check the piston spring for weakness or damage.

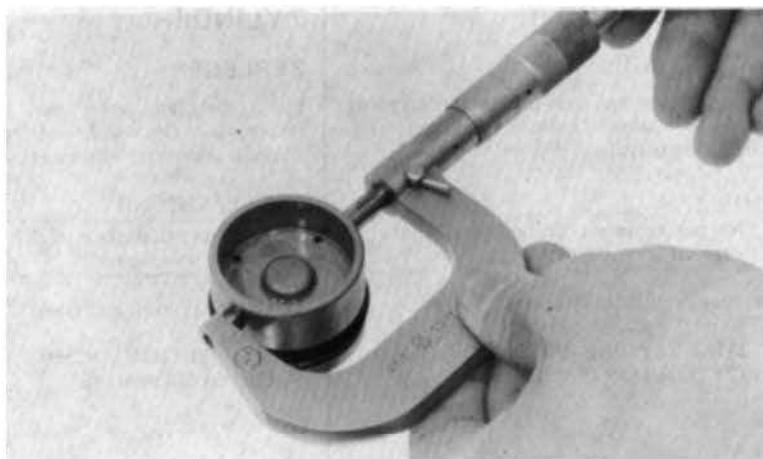




PISTON O.D. INSPECTION

Check the piston for scoring or scratches.
Measure the outside diameter of the piston with a micrometer.

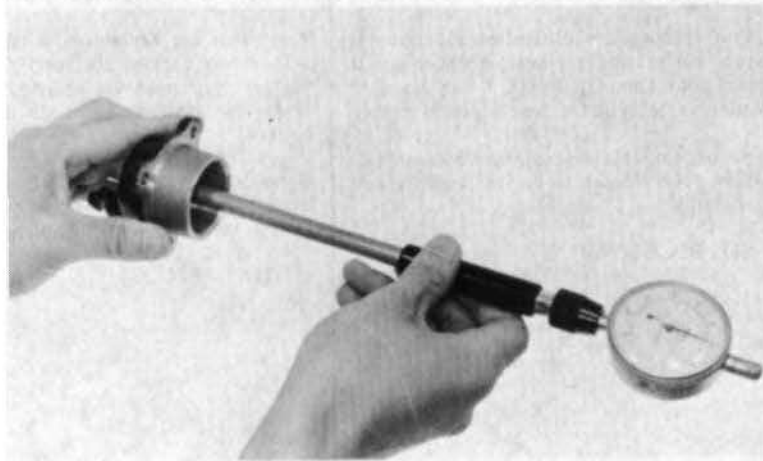
SERVICE LIMIT: 38.02 mm (1.497 in)



CYLINDER I.D. INSPECTION

Check the slave cylinder for scoring or scratches.
Measure the inside diameter of the cylinder bore.

SERVICE LIMIT: 38.18 mm (1.503 in)



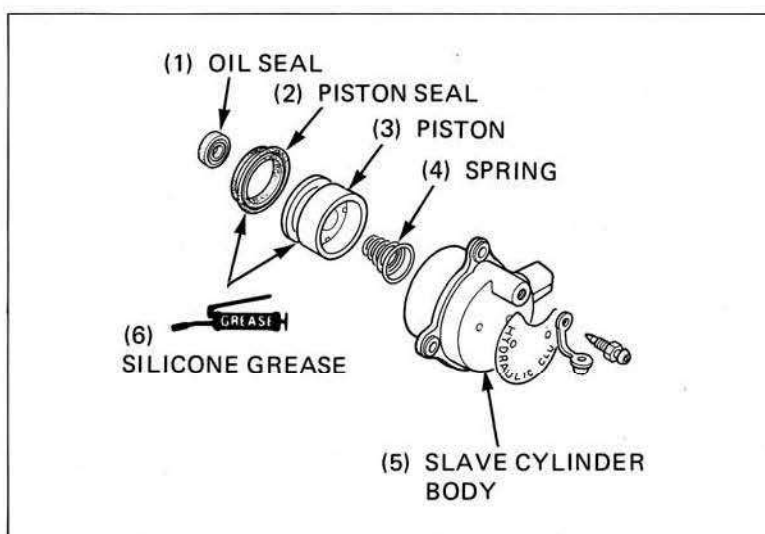
ASSEMBLY

Assemble the slave cylinder in the reverse order of disassembly. The oil seals must be replaced with new ones whenever they have been removed.

Lubricate the piston and piston seal with a medium grade of Hi-Temperature silicone grease or clutch fluid before assembly.

Be certain the piston seal is seated in the piston groove.

Place the piston in the cylinder with the oil seal end facing out.





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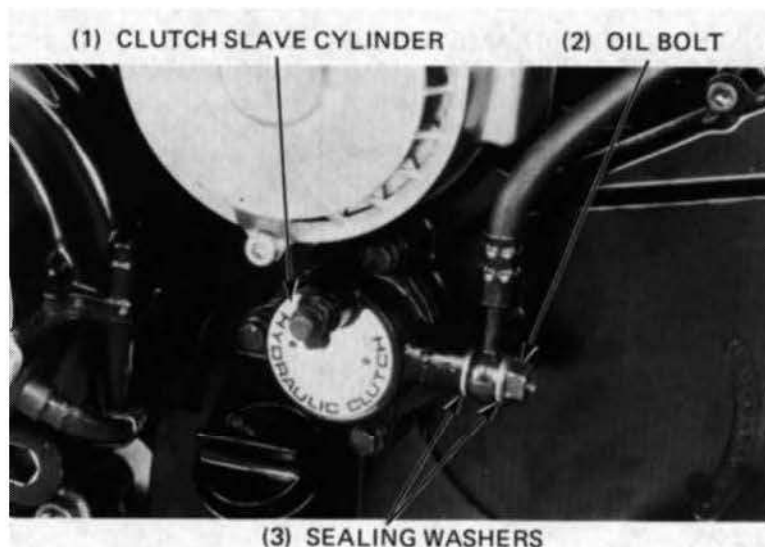
8. Clutch

Install the slave cylinder with a new gasket and tighten the slave cylinder bolts securely.

Connect the clutch hose with the oil bolt and two sealing washers, and tighten the oil bolt.

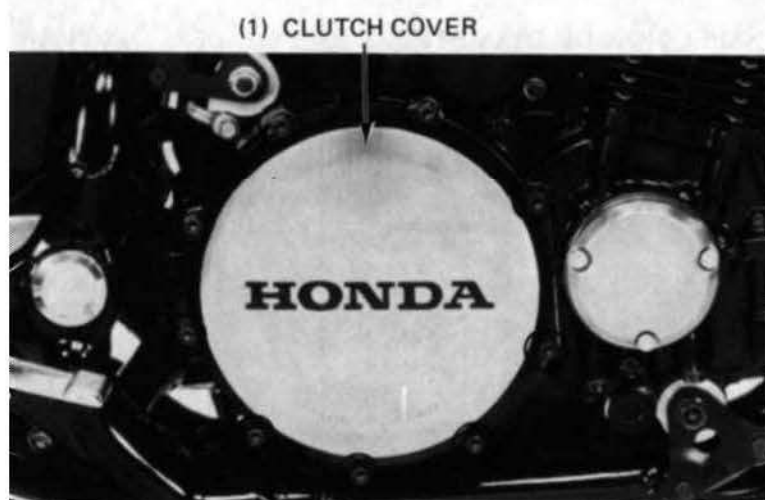
TORQUE. 25-30 Nm
(2.5-3.0 kg.m, 18-22 ft.lb)

Fill the clutch fluid reservoir and bleed the clutch system (page 8-3).



CLUTCH DISASSEMBLY

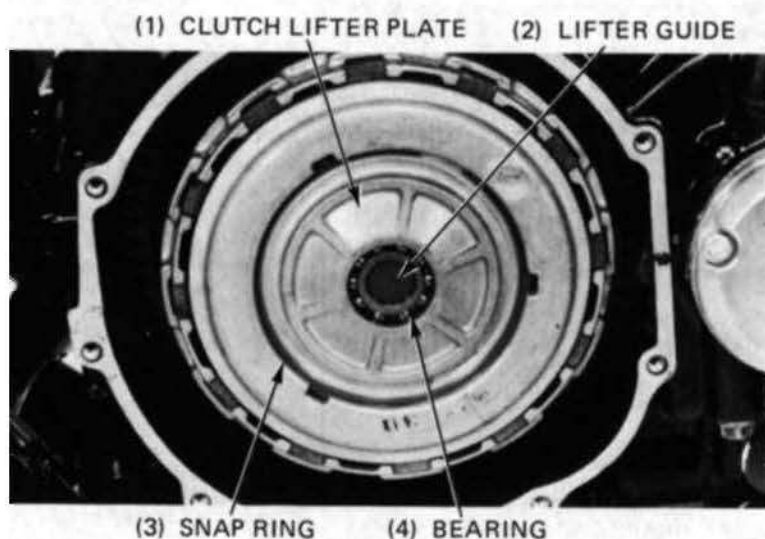
Drain the engine oil (page 2-3).
Remove the clutch cover and gasket.



Remove the snap ring, clutch lifter plate, bearing, lifter guide and lifter rod.

CAUTION

To prevent the clutch system from air contamination, squeeze the clutch lever immediately after removing the clutch lifter plate, and tie the lever to the handle grip with a string.





8. Clutch

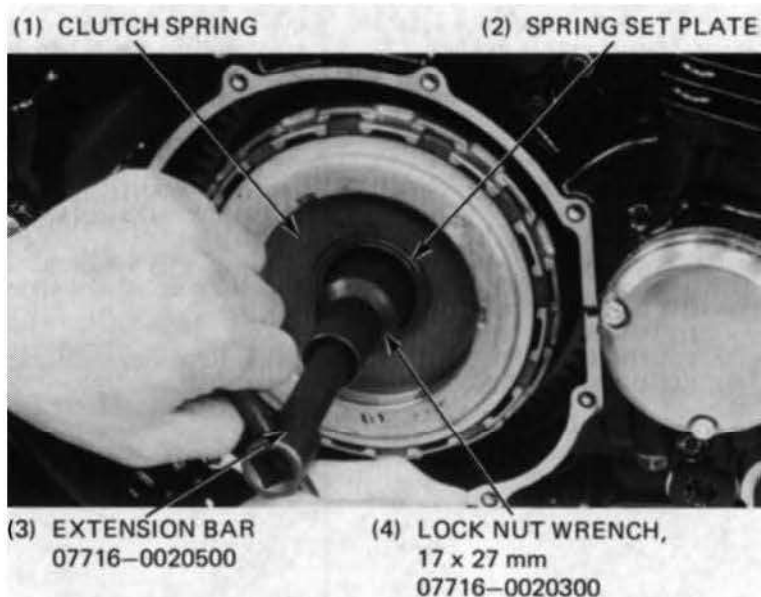
Shift the transmission into 6th gear and apply the rear brake.

NOTE:

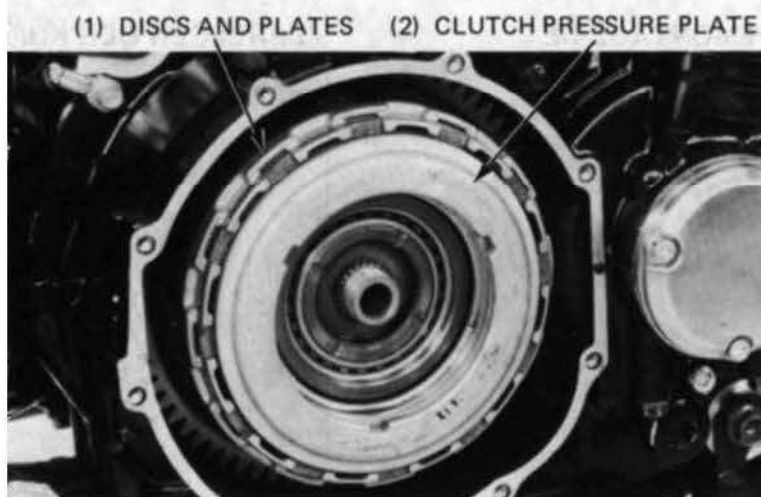
If the engine is not in the frame, shift the transmission into gear and use the universal holder (07725-0030000) to hold the drive sprocket.

Remove the lock nut.

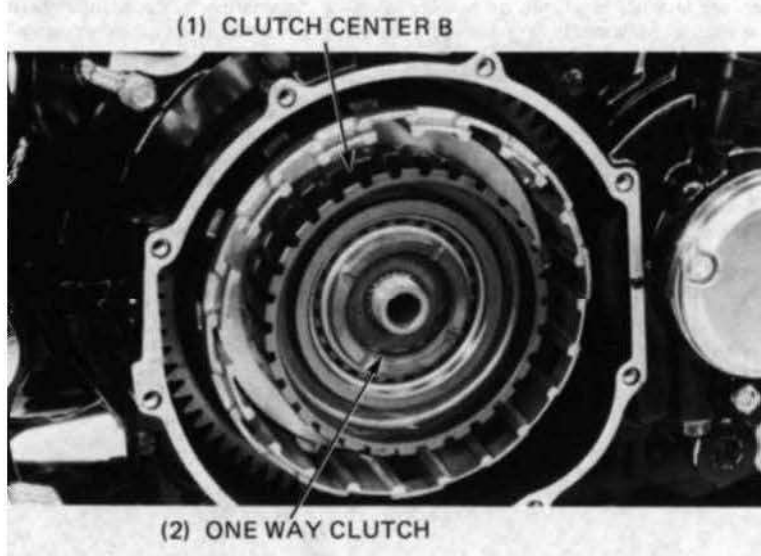
Remove the clutch spring set plate, clutch spring and two washers.



Remove the clutch pressure plate.
Remove the clutch plates and discs.

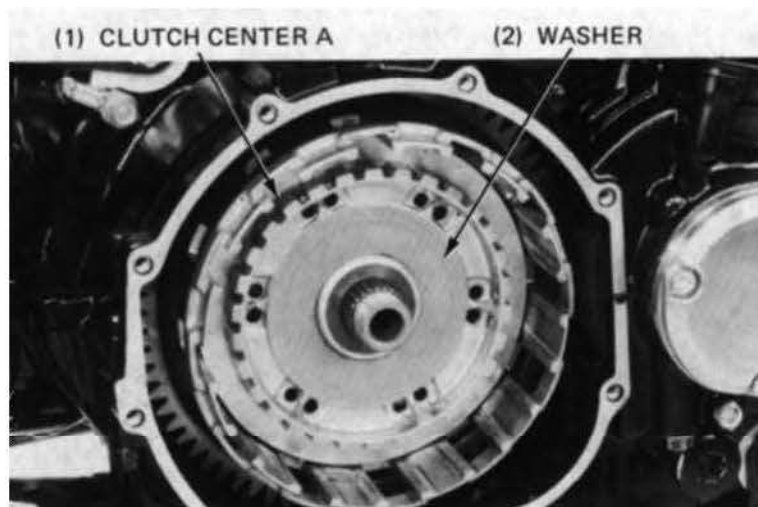


Remove clutch centre B and the one-way clutch as an assembly.

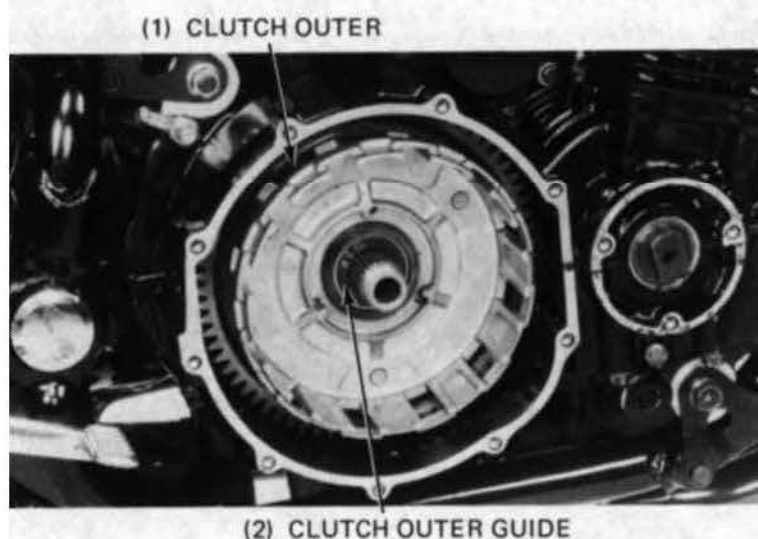




Remove clutch centre A and washer.



Remove the right crankshaft cover and turn the crankshaft clockwise until the No. 4 crank weight is positioned at BDC. Pull out the clutch outer guide and remove the clutch outer.



INSPECTION

CLUTCH SPRING

Measure the height of the clutch spring.

SERVICE LIMIT: 3.9 mm (0.15 in)

Replace the spring if it is shorter than the service limit.



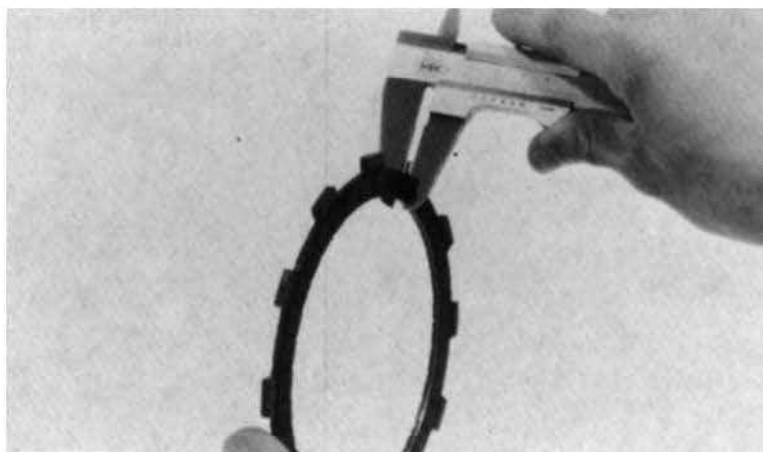


CLUTCH DISC

Replace the clutch discs if they show signs of scoring or discoloration. Measure the thickness of each disc.

SERVICE LIMIT: 3.1 mm (0.12 in)

Replace any discs that are thinner than the service limit.



CLUTCH PLATE

Check for plate warpage on a surface plate, using a feeler gauge.

SERVICE LIMIT: 0.30 mm (0.012 in)



ONE WAY CLUTCH INSPECTION

Inspect the one way clutch for smooth operation. Check the rollers for excessive wear.



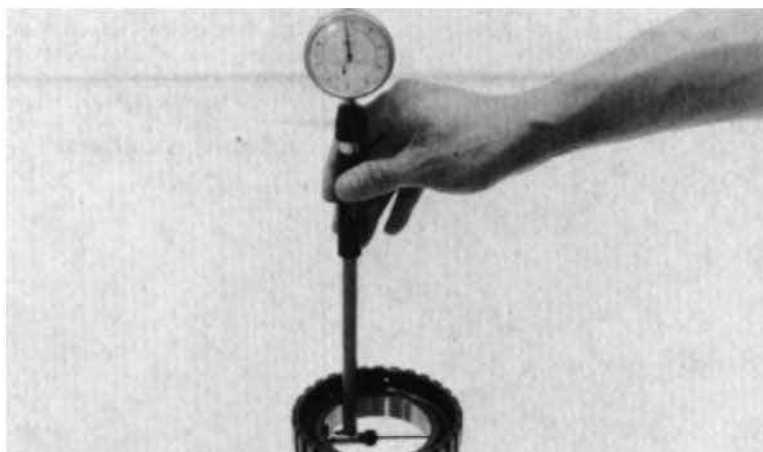


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8. Clutch

Measure the I.D. of clutch centre B.

SERVICE LIMIT: 74.50 mm (2.933 In)



Measure the O.D. of the one way clutch inner.

SERVICE LIMIT: 57.74 mm (2.273 In)



CLUTCH OUTER

Check the slots in the clutch outer for nicks, cuts or indentations made by the friction discs.

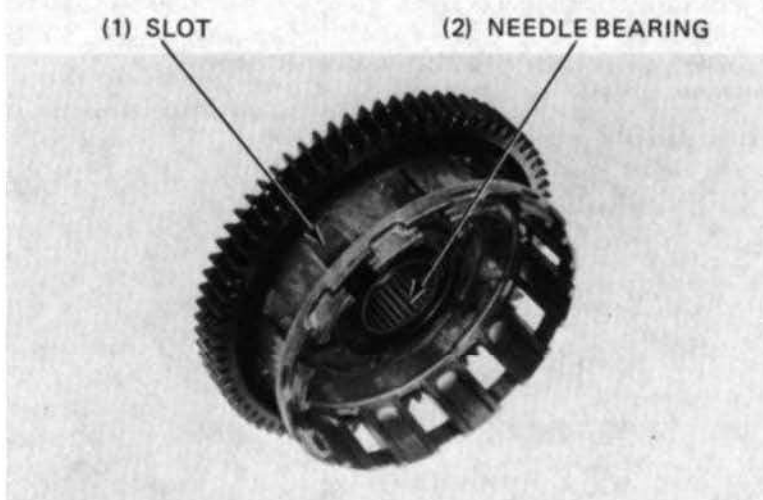
Check the clutch outer needle bearing for damage or excessive play.

If the needle bearing is difficult to remove from the clutch housing, use the following tools:

Driver: 07749-0010000

Attachment, 37 x 40 mm: 07746-0010200

Pilot, 35 mm: 07746-0040800

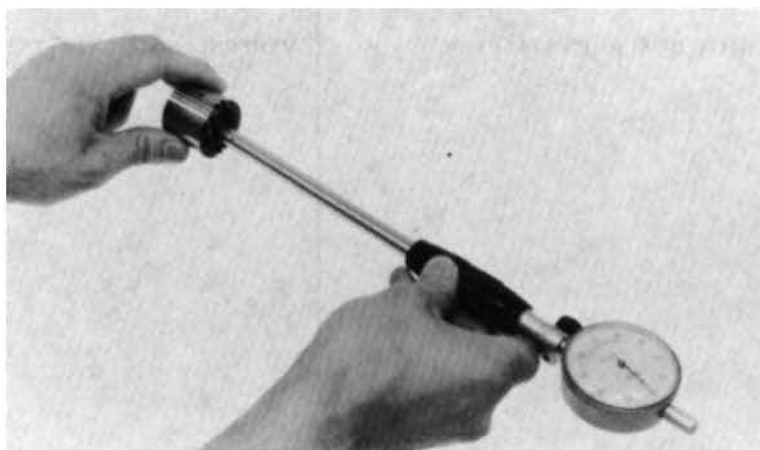




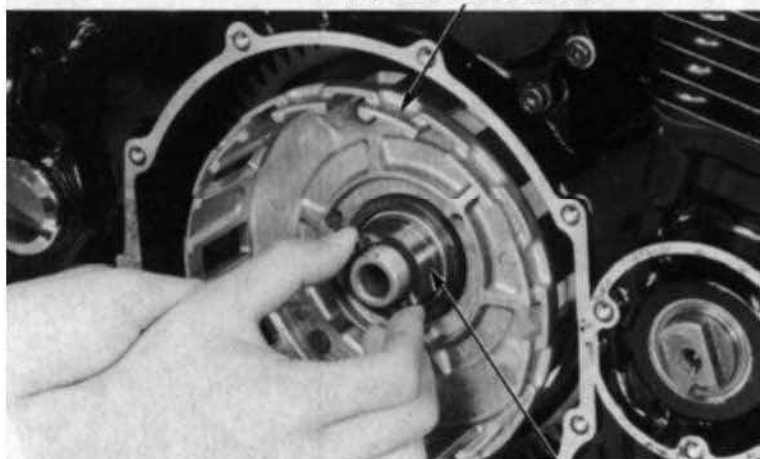
CLUTCH OUTER GUIDE

Measure the I.D. of the clutch outer guide.

SERVICE LIMIT: 25.08 mm (0.987 in)



(1) CLUTCH OUTER



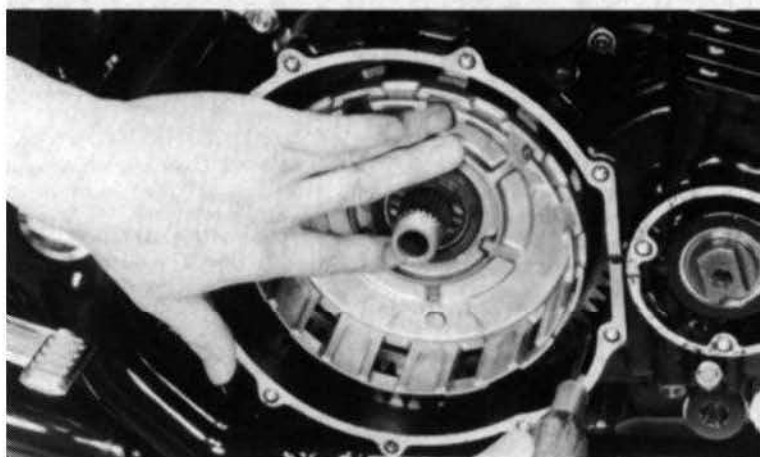
(2) CLUTCH OUTER GUIDE

CLUTCH ASSEMBLY

Install the needle bearing into the clutch outer. Install the clutch outer over the mainshaft.

Install the clutch outer guide between the mainshaft and clutch outer and push it in until it stops.

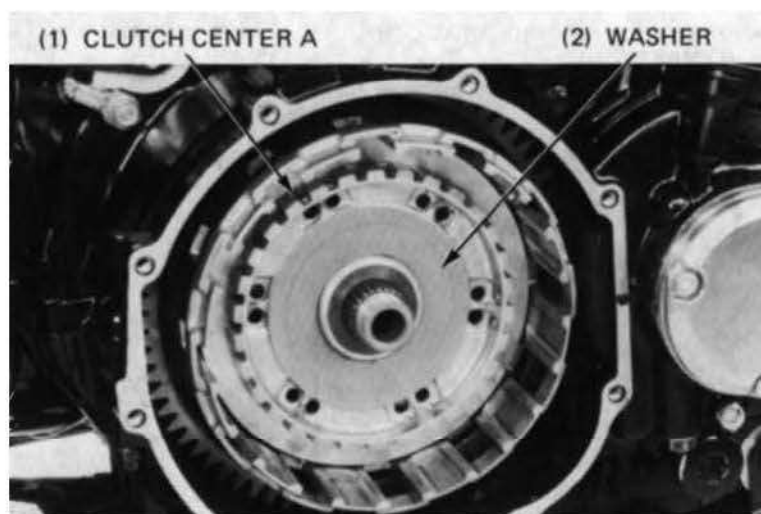
Push the clutch outer in while moving the primary driven gears with a screwdriver, then further push it in while moving the oil pump driven sprocket with the screwdriver to fit the pins on the drive sprocket into the holes in the clutch outer.





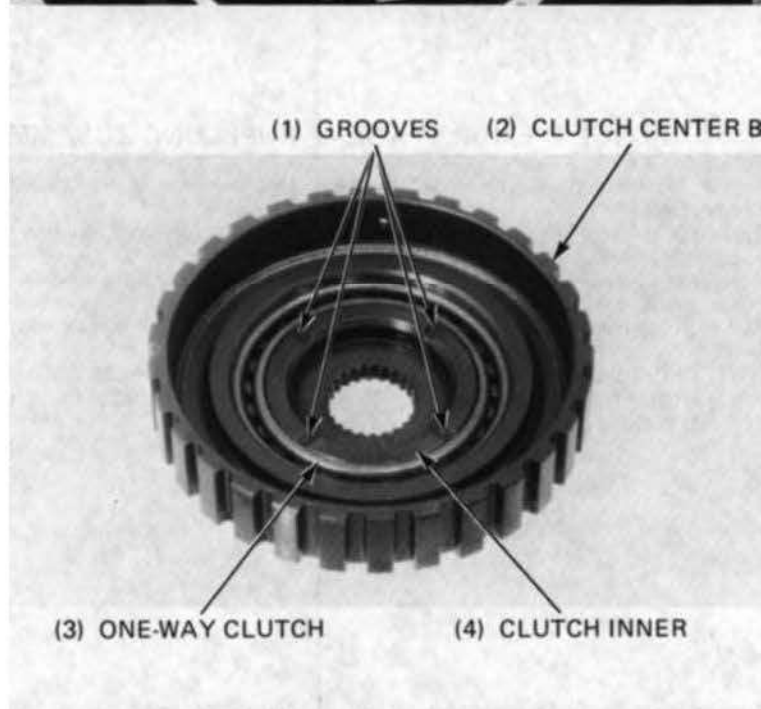
8. Clutch

Install clutch centre A and the washer.



Place the clutch centre B with the grooved side facing down.

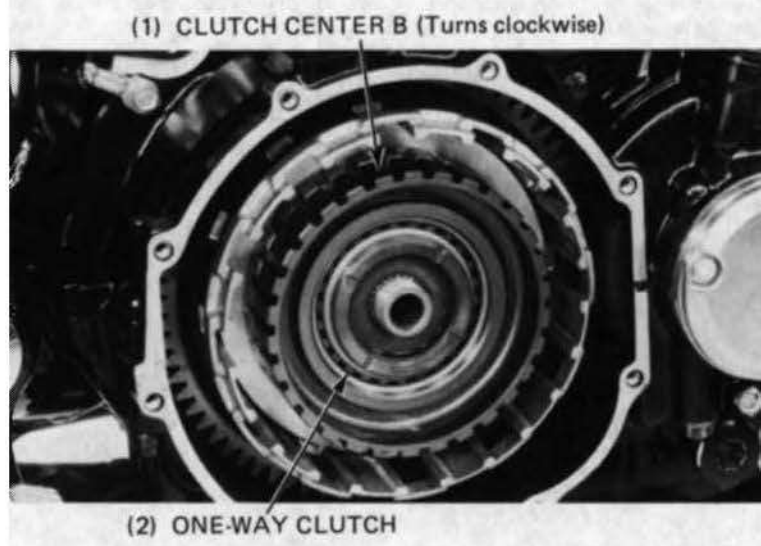
Install the one-way clutch into the clutch centre B with its flanged cage facing up. Install the clutch inner into the one-way clutch with its grooves facing up. Turn it counter clockwise as you install it.



Install the one-way clutch/clutch centre B assembly over the mainshaft.

NOTE:

Make sure the one way clutch assembly is installed correctly by turning the clutch centre B. The clutch centre should turn clockwise freely and should not turn counter clockwise.

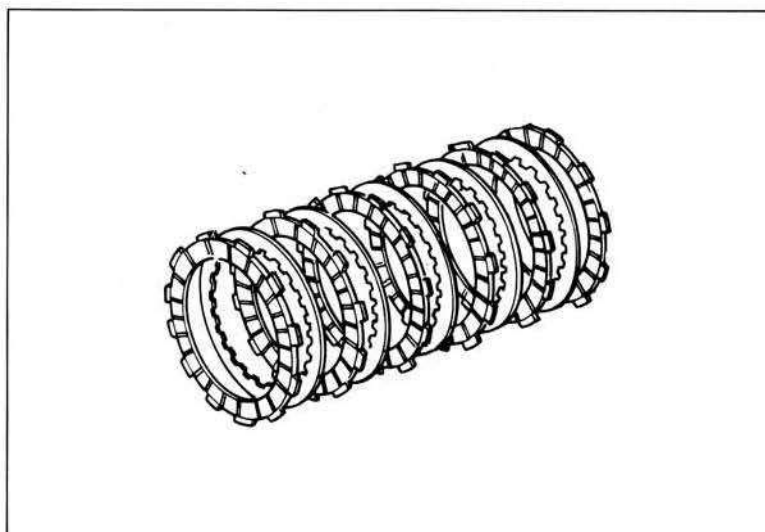




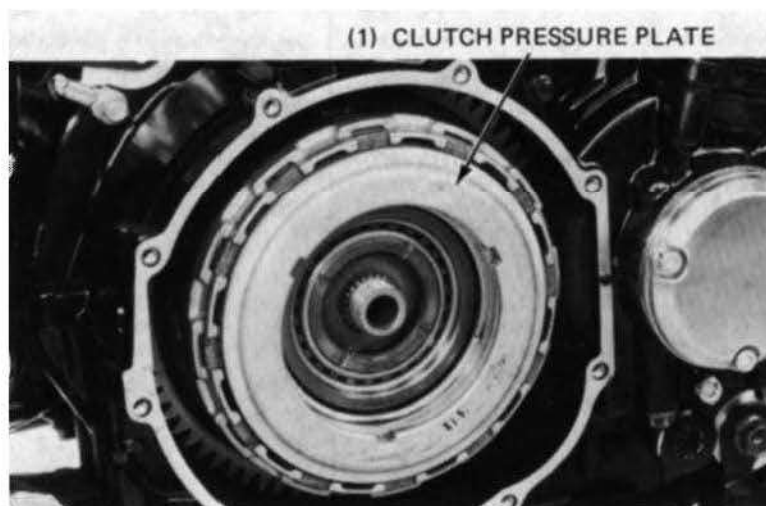
HONDA CBX750F

8. Clutch

Coat the discs and plates with clean engine oil, and install them.



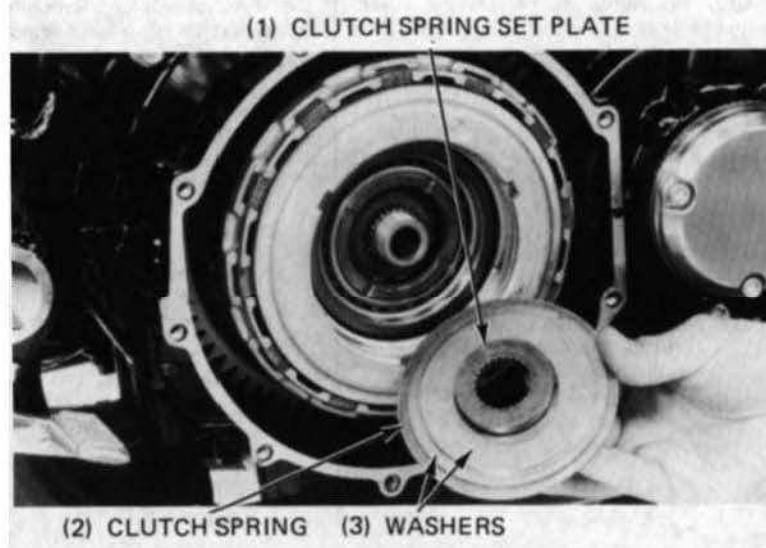
Install the clutch pressure plate.



Install the clutch spring set plate, clutch spring, and washers.

NOTE:

Install the clutch spring with the dished face towards the inside.





HONDA CBX750F

8. Clutch

Place the transmission in 6th gear.
Apply thread lock agent to the threads of the lock nut and mainshaft.
Install and tighten the lock nut.

NOTE:

If servicing the clutch with the engine out of the frame, shift the transmission into gear and hold the drive sprocket with the HOLDER 07725-0030000.

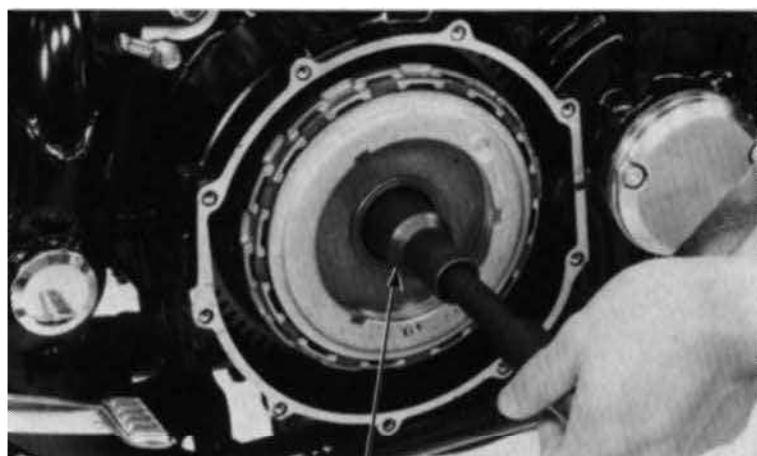
TORQUE: 75-85 Nm
(7.5-8.5 kg.m, 54-61 ft.lb)

Install the clutch lifter rod. Install the clutch lifter plate, lifter guide and bearing.

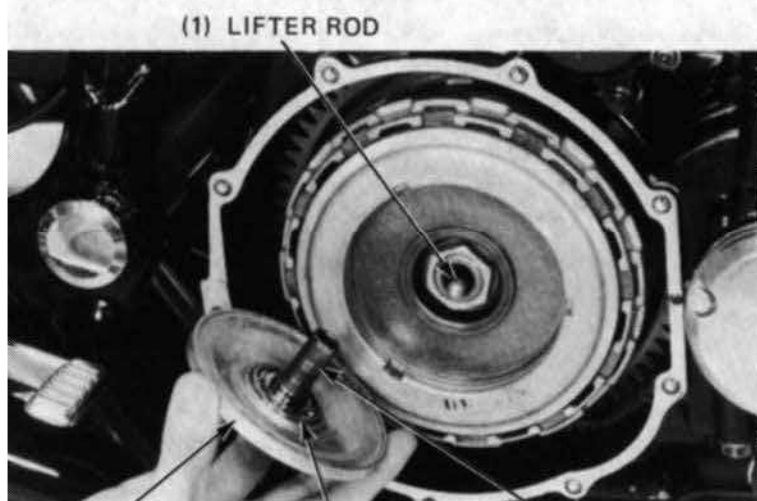
NOTE:

Before installing the lifter plate, release the clutch lever by removing the string.

Install the snap ring.

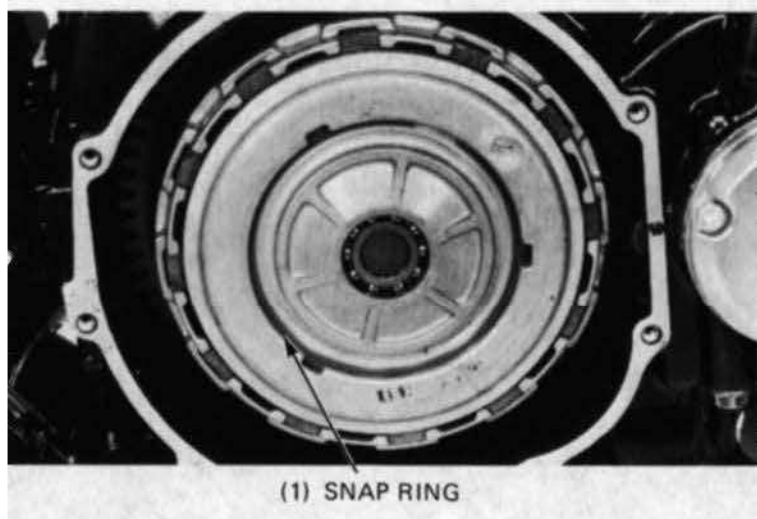


(1) LOCK NUT WRENCH 17 x 27 mm
07716-0020300



(1) LIFTER ROD

(2) LIFTER PLATE (3) BEARING (4) LIFTER GUIDE



(1) SNAP RING



HONDA

CBX750F

8. Clutch

Install the clutch cover with a new gasket.
Fill the crankcase with the recommended oil
(page 2-3).

